Phase 4: Magnetostatics

Steady Currents and Magnetic Fields

Master Study Guide & Step-by-Step Derivations

Contents

1 The Continuity Equation and Steady Currents

Before we can calculate magnetic fields, we must establish the physical rules that govern the currents creating them. The most fundamental rule of electricity is the **Conservation of Charge**. Charge cannot be created from nothing, nor can it be destroyed.

1.1 Deriving the Divergence of J

The Concept: Imagine a sealed, imaginary 3D balloon (a closed surface S bounding a volume V). If you observe a net electrical current flowing *out* through the skin of this balloon, the total amount of charge trapped *inside* the balloon must be decreasing.

Let $\mathbf{J}$ be the Volume Current Density (measured in Amperes/m^2). The total outward current I_{out} passing through the closed surface S is mathematically defined as the surface integral of $\mathbf{J}$:

$$I_{\text{out}} = \oint_S \mathbf{J} \cdot d\mathbf{S} \tag{1}$$

By the strict law of conservation of charge, this outward current must equal the negative rate of change of the total internal charge Q_{in} with respect to time:

$$\oint_S \mathbf{J} \cdot d\mathbf{S} = -\frac{dQ_{\text{in}}}{dt} \tag{2}$$

We know that the total internal charge Q_{in} can be expanded as the volume integral of the continuous volume charge density ρ_v (in Coulombs/m^3):

$$Q_{\text{in}} = \iiint_V \rho_v \, dv \tag{3}$$

Substitute this volume integral back into Equation 2:

$$\oint_S \mathbf{J} \cdot d\mathbf{S} = -\frac{d}{dt} \left(\iiint_V \rho_v \, dv \right) \tag{4}$$

Assuming our imaginary balloon surface S is completely stationary (it is not expanding or moving), we can safely slide the time derivative inside the spatial integral, turning it into a partial derivative:

$$\oint_S \mathbf{J} \cdot d\mathbf{S} = \iiint_V \left(-\frac{\partial \rho_v}{\partial t} \right) dv \tag{5}$$

Now, we deploy the **Divergence Theorem** exactly on the left-hand side. The Divergence Theorem allows us to mathematically convert the closed surface integral of a vector field into a volume integral of the spatial divergence of that field:

$$\oint_S \mathbf{J} \cdot d\mathbf{S} = \iiint_V (\nabla \cdot \mathbf{J})\, dv \tag{6}$$

Substitute this divergence conversion back into our equation:

$$\iiint_V (\nabla \cdot \mathbf{J})\, dv = \iiint_V \left(-\frac{\partial \rho_v}{\partial t} \right) dv \tag{7}$$

Group everything under a single volume integral:

$$\iiint_V \left[\nabla \cdot \mathbf{J} + \frac{\partial \rho_v}{\partial t} \right] dv = 0 \tag{8}$$

Because the initial physical choice of the bounded volume V was entirely arbitrary, this relationship must hold true for any conceivable volume in the universe. The only mathematical way this can happen is if the internal integrand itself evaluates identically to zero everywhere in space.

$$\nabla \cdot \mathbf{J} + \frac{\partial \rho_v}{\partial t} = 0 \tag{9}$$

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t} \tag{10}$$

Equation 10 is the universally valid **Continuity Equation**. It states that the divergence (the outward "spraying") of current density is directly tied to the depletion rate of local charge density.

1.2 The Steady Current Condition (PYQ Focus)

The university exam frequently asks you to "Show that $\nabla \cdot \mathbf{J} = 0$". This specific mathematical condition is only true under the strict assumption of **Steady Currents** (the realm of Magnetostatics).

Logical Deduction: A steady current (DC current) is defined physically as a continuous, constant flow of charge that does not alter or fluctuate with respect to time. By definition, in a steady-state system, charge is never allowed to pile up, accumulate, or deplete at any localized point in space. It just flows through smoothly.

Therefore, the local volume charge density ρ_v remains absolutely constant over time:

$$\frac{\partial \rho_v}{\partial t} = 0 \quad \text{(Strict Condition for Steady Currents)} \tag{11}$$

When we substitute this rigid steady-state condition back into our universally derived Continuity Equation:

$$\nabla \cdot \mathbf{J} = -(0) \tag{12}$$

$$\nabla \cdot \mathbf{J} = 0 \tag{13}$$

Final Result & Physical Conclusion:

The expression $\nabla \cdot \mathbf{J} = 0$ proves mathematically that steady DC current fields are perfectly **solenoidal**. Whatever amount of current flows into an arbitrary junction or volume must exactly equal the current flowing out of it. This provides the fundamental theoretical proof for **Kirchhoff's Current Law (KCL)**.

2 Ampere's Circuital Law

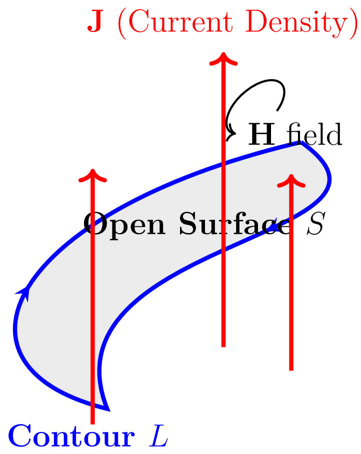
2.1 Establishing Curl H = J (PYQ Focus)

The Concept: In electrostatics, Gauss's Law relates electric flux to enclosed charge. In magnetostatics, **Ampere's Circuital Law** relates magnetic circulation to enclosed current.

Ampere discovered empirically that the line integral (circulation) of the magnetic field intensity vector $\mathbf{H}$ around any closed, continuous loop L is exactly equal to the total net direct current I_{enc} physically enclosed by and piercing through that loop.

$$\oint_L \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}} \tag{14}$$

TikZ Diagram: Ampere's Law



Step-by-Step Derivation:

Step 1: Expand the Enclosed Current

The total enclosed electrical current I_{enc} is formally defined as the open surface integral of the current density vector field $\mathbf{J}$ penetrating the arbitrary surface area S bounded by the contour L:

$$I_{\text{enc}} = \iint_S \mathbf{J} \cdot d\mathbf{S} \tag{15}$$

We mathematically substitute Equation 15 directly into Ampere's Law (Equation 14):

$$\oint_L \mathbf{H} \cdot d\mathbf{l} = \iint_S \mathbf{J} \cdot d\mathbf{S} \tag{16}$$

Step 2: Apply Stokes' Theorem

We must apply **Stokes' Theorem** directly to the left-hand side of this integrated equation. Stokes' Theorem seamlessly converts the closed line integral of a vector field into an open surface integral of the mathematical *curl* of that exact vector field over the bounded area:

$$\oint_L \mathbf{H} \cdot d\mathbf{l} = \iint_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S} \tag{17}$$

Step 3: Synthesis

We substitute this curl conversion back into our combined Ampere's Law equation (Equation 16):

$$\iint_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S} = \iint_S \mathbf{J} \cdot d\mathbf{S} \tag{18}$$

Rearranging all terms together under a single unified integral bracket:

$$\iint_S [(\nabla \times \mathbf{H}) - \mathbf{J}] \cdot d\mathbf{S} = 0 \tag{19}$$

Because the initial physical choice of the open bounding surface S was completely and entirely arbitrary, this integral relationship must hold rigorously true for any conceivable surface shape drawn in the universe. The only way a surface integral can evaluate to zero for absolutely all possible surface topologies is if the internal integrand itself evaluates identically to zero everywhere in space.

$$(\nabla \times \mathbf{H}) - \mathbf{J} = \mathbf{0} \tag{20}$$

$$\nabla \times \mathbf{H} = \mathbf{J} \tag{21}$$

Final Result & Physical Conclusion:

The derived point-form equation $\nabla \times \mathbf{H} = \mathbf{J}$ proves that a moving stream of electrical charge (current density $\mathbf{J}$) acts as the direct, fundamental physical source that generates and "curls" the magnetic field ($\mathbf{H}$) into continuous, closed loops encircling the current path.

3 Magnetic Vector Potential & Biot-Savart Law

In electrostatics, we simplify the calculation of electric fields by defining a scalar potential function V such that $\mathbf{E} = -\nabla V$. This is possible because the static electric field is conservative ($\nabla \times \mathbf{E} = \mathbf{0}$).

However, in magnetostatics, the magnetic field is *not* conservative where currents exist, because $\nabla \times \mathbf{H} = \mathbf{J}$. Therefore, we cannot broadly express $\mathbf{B}$ as the gradient of a scalar potential.

3.1 Defining the Vector Potential (A)

Instead, we rely on the fundamental Maxwell equation stating that magnetic fields have zero divergence (no magnetic monopoles):

$$\nabla \cdot \mathbf{B} = 0 \tag{22}$$

A fundamental theorem of vector calculus states that if the divergence of a vector field is identically zero, that field can always be mathematically expressed as the curl of another vector field. Therefore, we define the **Magnetic Vector Potential**, denoted as $\mathbf{A}$ (measured in Webers/meter), such that:

$$\mathbf{B} = \nabla \times \mathbf{A} \tag{23}$$

The Coulomb Gauge:
Specifying the curl of a vector field does not uniquely define it. We can add the gradient of any arbitrary scalar function to $\mathbf{A}$ without altering the resulting $\mathbf{B}$ field, since the curl of a gradient is always zero ($\nabla \times \nabla\psi = \mathbf{0}$). To uniquely lock down the behavior of $\mathbf{A}$, we must mathematically specify its divergence. For static fields, we universally choose the **Coulomb Gauge**:

$$\nabla \cdot \mathbf{A} = 0 \tag{24}$$

3.2 Deducing the Biot-Savart Law (PYQ 2022)

Step 1: Poisson's Equation for A
We substitute our definition of the vector potential ($\mathbf{B} = \nabla \times \mathbf{A}$) into the differential form of Ampere's Law ($\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$):

$$\nabla \times (\nabla \times \mathbf{A}) = \mu_0 \mathbf{J} \tag{25}$$

We apply the standard vector calculus identity for the "curl of a curl":

$$\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} \tag{26}$$

Substitute this identity back into Ampere's Law:

$$\nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} = \mu_0 \mathbf{J} \tag{27}$$

Because we deliberately chose the Coulomb gauge ($\nabla \cdot \mathbf{A} = 0$), the first term vanishes entirely. This simplifies the complex coupled differential equations into a set of three independent Poisson's equations (one for each Cartesian vector component):

$$\nabla^2 \mathbf{A} = -\mu_0 \mathbf{J} \tag{28}$$

Step 2: The Integral Solution

We know from electrostatics that the scalar Poisson's equation $\nabla^2 V = -\frac{\rho_v}{\epsilon_0}$ possesses the general integral solution:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \iiint_{v'} \frac{\rho_v(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} dv' \tag{29}$$

By direct mathematical analogy, the vector Poisson's equation (Equation 28) possesses an identical integral solution structure:

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \iiint_{v'} \frac{\mathbf{J}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} dv' \tag{30}$$

Where:

- $\mathbf{r}$ is the unprimed position vector of the observation point.
- $\mathbf{r}'$ is the primed position vector of the source current element.
- $R = |\mathbf{r} - \mathbf{r}'|$ is the scalar distance between the source and observation points.

Step 3: Taking the Curl

To find the magnetic flux density $\mathbf{B}$, we take the curl of this integral solution:

$$\mathbf{B} = \nabla \times \mathbf{A} = \nabla \times \left(\frac{\mu_0}{4\pi} \iiint_{v'} \frac{\mathbf{J}(\mathbf{r}')}{R} dv' \right) \tag{31}$$

Because the curl operator ∇ involves spatial derivatives strictly with respect to the *observation* coordinates (x, y, z), and the volume integral is strictly over the *source* coordinates (x', y', z'), we can confidently move the curl operator inside the integral:

$$\mathbf{B} = \frac{\mu_0}{4\pi} \iiint_{v'} \nabla \times \left(\frac{\mathbf{J}(\mathbf{r}')}{R} \right) dv' \tag{32}$$

Step 4: Vector Product Rule

We must apply the vector product rule for the curl of a scalar $(1/R)$ multiplied by a vector $(\mathbf{J})$:

$$\nabla \times (\mathbf{fC}) = f(\nabla \times \mathbf{C}) - \mathbf{C} \times (\nabla f) \tag{33}$$

Applying this to our specific variables:

$$\nabla \times \left(\frac{\mathbf{J}}{R} \right) = \frac{1}{R}(\nabla \times \mathbf{J}) - \mathbf{J} \times \nabla \left(\frac{1}{R} \right) \tag{34}$$

Because $\mathbf{J}$ is purely a function of the primed source coordinates, its curl with respect to the unprimed observation coordinates is zero $(\nabla \times \mathbf{J} = \mathbf{0})$. The first term vanishes.

The gradient of the scalar distance function $1/R$ with respect to the unprimed observation coordinates evaluates to:

$$\nabla \left(\frac{1}{R} \right) = -\frac{\mathbf{R}}{R^3} = -\frac{\hat{\mathbf{a}}_R}{R^2} \tag{35}$$

Substituting this gradient back:

$$\nabla \times \left(\frac{\mathbf{J}}{R} \right) = -\mathbf{J} \times \left(-\frac{\hat{\mathbf{a}}_R}{R^2} \right) \tag{36}$$

$$= \frac{\mathbf{J} \times \hat{\mathbf{a}}_R}{R^2} \tag{37}$$

Step 5: The Final Biot-Savart Equation

We re-insert this cross product back into our macroscopic volume integral:

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \iiint_{v'} \frac{\mathbf{J}(\mathbf{r}') \times \hat{\mathbf{a}}_R}{R^2} dv' \tag{38}$$

For a thin, filamentary wire carrying a total current I, the volume current density element simplifies: $\mathbf{J} dv' \longrightarrow I d\mathbf{l}'$. Making this substitution yields the final formula:

Final Result & Physical Conclusion:

The Biot-Savart Law:

$$\mathbf{B} = \frac{\mu_0 I}{4\pi} \oint_L \frac{d\mathbf{l}' \times \hat{\mathbf{a}}_R}{R^2} \tag{39}$$

Where $d\mathbf{l}'$ is the differential length vector pointing in the direction of current flow, and $\hat{\mathbf{a}}_R$ is the unit vector pointing directly from the source element to the observation point.

4 Biot-Savart Application: Square Loop (PYQ 2023)

Problem Statement:

A square loop measuring 2 m by 2 m carries a 7.5 A steady current. The loop is in the xz plane, the origin coinciding with a corner of the square. Using Biot-Savart law compute the B-field at a point on the y-axis 0.35 m from the origin in air material.

4.1 General Formula: B-Field of a Finite Straight Wire

Before tackling the full square, we must derive the field of a single finite straight wire segment. Consider a straight wire carrying current I along the x-axis from x_1 to x_2. We want the field at an observation point P located at a perpendicular distance y_0 on the y-axis.

By the Biot-Savart law:

$$d\mathbf{B} = \frac{\mu_0 I}{4\pi} \frac{d\mathbf{l} \times \mathbf{R}}{|\mathbf{R}|^3} \tag{40}$$

Define the Vectors:

- Source element vector: $d\mathbf{l} = dx\hat{\mathbf{i}}$.
- Source position: $\mathbf{r}' = x\hat{\mathbf{i}}$.
- Observation position: $\mathbf{r} = y_0\hat{\mathbf{j}}$.
- Distance Vector $\mathbf{R} = \mathbf{r} - \mathbf{r}' = -x\hat{\mathbf{i}} + y_0\hat{\mathbf{j}}$.

Calculate the Cross Product:

$$\begin{aligned} d\mathbf{l} \times \mathbf{R} &= (dx\hat{\mathbf{i}}) \times (-x\hat{\mathbf{i}} + y_0\hat{\mathbf{j}}) && (41)\\ &= -xdx(\hat{\mathbf{i}} \times \hat{\mathbf{i}}) + y_0 dx(\hat{\mathbf{i}} \times \hat{\mathbf{j}}) && (42)\\ &= 0 + y_0 dx(\hat{\mathbf{k}}) = y_0 dx\hat{\mathbf{k}} && (43) \end{aligned}$$

Set up the Integral: The magnitude is $|\mathbf{R}|^3 = (x^2 + y_0^2)^{3/2}$.

$$\mathbf{B}_{\text{wire}} = \frac{\mu_0 I}{4\pi} \int_{x_1}^{x_2} \frac{y_0}{(x^2 + y_0^2)^{3/2}} dx\, \hat{\mathbf{k}} \tag{44}$$

Using the standard trigonometric substitution $x = y_0 \tan\theta$, the integral evaluates to:

$$\int \frac{dx}{(x^2 + y_0^2)^{3/2}} = \frac{x}{y_0^2\sqrt{x^2 + y_0^2}} \tag{45}$$

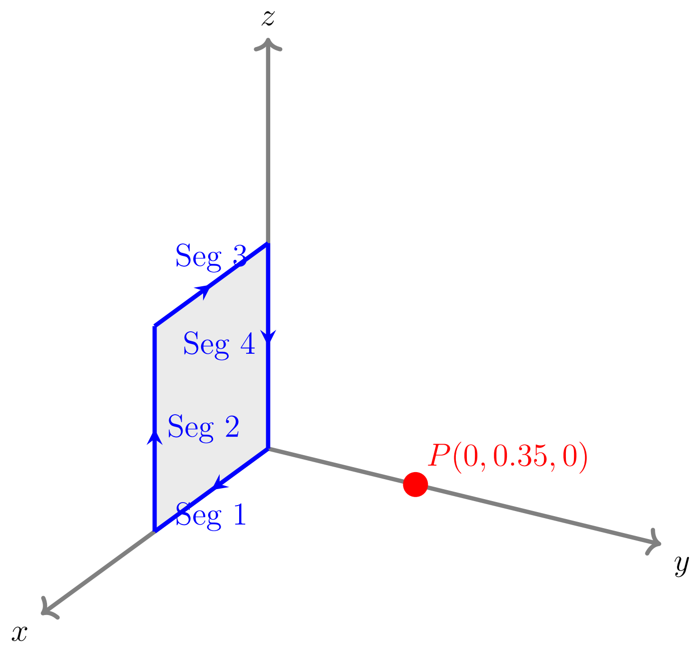
Substituting this back into Equation 44 yields our highly reusable finite wire formula:

$$\mathbf{B}_{\text{wire}} = \frac{\mu_0 I}{4\pi y_0} \left[\frac{x}{\sqrt{x^2 + y_0^2}} \right]_{x_1}^{x_2} \hat{\mathbf{k}} \tag{46}$$

4.2 Evaluating the Square Loop

We have four distinct wire segments forming the square in the xz-plane. The current $I = 7.5$ A flows counter-clockwise.

TikZ Diagram: 3D Square Loop Setup



Segment 1: From (0,0,0) to (2,0,0)
This segment lies entirely along the x-axis. We can plug its limits ($x_1 = 0, x_2 = 2$) directly into our derived formula (Equation 46), where the perpendicular distance to P is $y_0 = 0.35$:

$$\mathbf{B}_1 = \frac{\mu_0 I}{4\pi(0.35)} \left[\frac{2}{\sqrt{2^2 + 0.35^2}} - \frac{0}{\dots} \right] \hat{\mathbf{k}} \tag{47}$$

$$\mathbf{B}_1 = \frac{(4\pi \times 10^{-7})(7.5)}{4\pi(0.35)} \left(\frac{2}{\sqrt{4.1225}} \right) \hat{\mathbf{k}} \tag{48}$$

$$\mathbf{B}_1 = (2.14286 \times 10^{-6}) \times \left(\frac{2}{2.0304} \right) \hat{\mathbf{k}} \tag{49}$$

$$\mathbf{B}_1 = 2.1108 \times 10^{-6}\, \hat{\mathbf{k}} \text{ Tesla} \tag{50}$$

Segment 2: From (2,0,0) to (2,0,2)

This vertical segment extends from $z = 0$ to $z = 2$ at a constant offset $x = 2$. $d\mathbf{l} = dz\hat{\mathbf{k}}$. The distance vector is $\mathbf{R} = -2\hat{\mathbf{i}} + y_0\hat{\mathbf{j}} - z\hat{\mathbf{k}}$.

$$d\mathbf{l} \times \mathbf{R} = (dz\hat{\mathbf{k}}) \times (-2\hat{\mathbf{i}} + y_0\hat{\mathbf{j}} - z\hat{\mathbf{k}}) \tag{51}$$

$$= -2dz(\hat{\mathbf{j}}) + y_0dz(-\hat{\mathbf{i}}) = -y_0dz\hat{\mathbf{i}} - 2dz\hat{\mathbf{j}} \tag{52}$$

The denominator is $(2^2 + y_0^2 + z^2)^{3/2} = (4.1225 + z^2)^{3/2}$.

$$\mathbf{B}_2 = \frac{\mu_0 I}{4\pi}(-0.35\hat{\mathbf{i}} - 2\hat{\mathbf{j}}) \int_0^2 \frac{dz}{(4.1225 + z^2)^{3/2}} \tag{53}$$

$$= (7.5 \times 10^{-7})(-0.35\hat{\mathbf{i}} - 2\hat{\mathbf{j}}) \left[\frac{2}{4.1225\sqrt{4.1225 + 4}}\right] \tag{54}$$

$$= (7.5 \times 10^{-7})(-0.35\hat{\mathbf{i}} - 2\hat{\mathbf{j}}) \left(\frac{2}{11.749}\right) \tag{55}$$

$$= -0.0447 \times 10^{-6}\hat{\mathbf{i}} - 0.2553 \times 10^{-6}\hat{\mathbf{j}} \text{ Tesla} \tag{56}$$

Segment 3: From (2,0,2) to (0,0,2)

This segment extends backward from $x = 2$ to $x = 0$ at an offset $z = 2$. $d\mathbf{l} = dx\hat{\mathbf{i}}$. Distance vector $\mathbf{R} = -x\hat{\mathbf{i}} + y_0\hat{\mathbf{j}} - 2\hat{\mathbf{k}}$.

$$d\mathbf{l} \times \mathbf{R} = y_0dx(\hat{\mathbf{k}}) - 2dx(-\hat{\mathbf{j}}) = 2dx\hat{\mathbf{j}} + y_0dx\hat{\mathbf{k}} \tag{57}$$

Integrating from $x = 2$ to $x = 0$ (adds a negative sign to the evaluated bounds):

$$\mathbf{B}_3 = \frac{\mu_0 I}{4\pi} \int_2^0 \frac{2\hat{\mathbf{j}} + y_0\hat{\mathbf{k}}}{(x^2 + y_0^2 + 4)^{3/2}} dx \tag{58}$$

$$= -\frac{\mu_0 I}{4\pi}(2\hat{\mathbf{j}} + y_0\hat{\mathbf{k}}) \int_0^2 \frac{dx}{(x^2 + 4.1225)^{3/2}} \tag{59}$$

Notice the integral evaluates to the exact same scalar constant as Segment 2 (2/11.749).

$$\mathbf{B}_3 = -(7.5 \times 10^{-7})(2\hat{\mathbf{j}} + 0.35\hat{\mathbf{k}}) \left(\frac{2}{11.749}\right) \tag{60}$$

$$\mathbf{B}_3 = -0.2553 \times 10^{-6}\hat{\mathbf{j}} - 0.0447 \times 10^{-6}\hat{\mathbf{k}} \text{ Tesla} \tag{61}$$

Segment 4: From (0,0,2) to (0,0,0)

This segment extends downward from $z = 2$ to $z = 0$ along the z-axis. $d\mathbf{l} = dz\hat{\mathbf{k}}$. Distance vector $\mathbf{R} = y_0\hat{\mathbf{j}} - z\hat{\mathbf{k}}$.

$$d\mathbf{l} \times \mathbf{R} = y_0dz(-\hat{\mathbf{i}}) = -y_0dz\hat{\mathbf{i}} \tag{62}$$

Integrate from $z = 2$ to $z = 0$:

$$\mathbf{B}_4 = \frac{\mu_0 I}{4\pi} \int_2^0 \frac{-y_0 \hat{\mathbf{i}}}{(y_0^2 + z^2)^{3/2}} dz \tag{63}$$

$$= \frac{\mu_0 I y_0}{4\pi} \hat{\mathbf{i}} \int_0^2 \frac{dz}{(0.35^2 + z^2)^{3/2}} \tag{64}$$

Notice this integral is mathematically identical to Segment 1!

$$\mathbf{B}_4 = 2.1108 \times 10^{-6}\, \hat{\mathbf{i}} \text{ Tesla} \tag{65}$$

Total Superposition:

We perform a vector addition of all four components:

$$\mathbf{B}_{\text{total}} = \mathbf{B}_1 + \mathbf{B}_2 + \mathbf{B}_3 + \mathbf{B}_4 \tag{66}$$

$$B_x = 0 - 0.0447 + 0 + 2.1108 = 2.0661\,\mu\text{T} \tag{67}$$

$$B_y = 0 - 0.2553 - 0.2553 + 0 = -0.5106\,\mu\text{T} \tag{68}$$

$$B_z = 2.1108 + 0 - 0.0447 + 0 = 2.0661\,\mu\text{T} \tag{69}$$

Final Result & Physical Conclusion:

The total magnetic flux density $\mathbf{B}$ at point $P(0, 0.35, 0)$ is:

$$\mathbf{B}_{\text{total}} \approx 2.066\hat{\mathbf{i}} - 0.511\hat{\mathbf{j}} + 2.066\hat{\mathbf{k}} \quad [\mu\text{T}]$$

5 Magnetic Boundary Conditions

What happens when a magnetic field passes from one material (like Air, μ_1) into a highly permeable material (like an Iron Core, μ_2)? We must establish rules for the Normal (B_n) and Tangential (H_t) components.

5.1 The Normal Component (B_n)

Because there are no magnetic monopoles, the magnetic field is purely solenoidal.

$$\nabla \cdot \mathbf{B} = 0 \implies \oint_S \mathbf{B} \cdot d\mathbf{S} = 0 \tag{70}$$

Imagine placing a microscopic, flat cylindrical Gaussian pillbox halfway through the boundary. As the height of the pillbox shrinks to zero ($\Delta h \to 0$), the curved side area vanishes. Flux only enters the bottom face and exits the top face.

$$\int_{\text{top}} \mathbf{B}_1 \cdot d\mathbf{S} + \int_{\text{bottom}} \mathbf{B}_2 \cdot d\mathbf{S} = 0 \tag{71}$$

$$B_{n1}\Delta S - B_{n2}\Delta S = 0 \tag{72}$$

$$B_{n1} = B_{n2} \tag{73}$$

Conclusion: The normal component of the magnetic flux density ($\mathbf{B}$) is perfectly continuous across a boundary. Since $\mathbf{B} = \mu\mathbf{H}$, the normal component of the magnetic field intensity ($\mathbf{H}$) is physically discontinuous ($\mu_1 H_{n1} = \mu_2 H_{n2}$).

5.2 The Tangential Component (H_t)

We use Ampere's Law for the tangential component:

$$\oint_L \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}} \tag{74}$$

Imagine drawing a microscopic rectangular loop straddling the boundary, with width Δw and height Δh. If a sheet of free surface current $\mathbf{K}$ (Amperes/meter) is flowing exactly along the boundary interface, the loop encloses a total current of $I_{\text{enc}} = K\Delta w$.

As we shrink the height of the loop to zero ($\Delta h \to 0$), the line integrals along the vertical paths vanish. We are left with the horizontal path moving right through Medium

1, and backward through Medium 2:

$$H_{t1}\Delta w - H_{t2}\Delta w = K\Delta w \tag{75}$$

$$H_{t1} - H_{t2} = K \tag{76}$$

Conclusion: The tangential component of the magnetic field intensity ($\mathbf{H}$) is discontinuous exactly by the magnitude of the surface current density ($\mathbf{K}$) flowing on the boundary.

If the materials are standard finite conductors or insulators, an infinitely thin surface current cannot physically exist ($\mathbf{K} = 0$). In this standard case:

$$H_{t1} = H_{t2} \implies \frac{B_{t1}}{\mu_1} = \frac{B_{t2}}{\mu_2} \tag{77}$$

6 Phase 4 Summary and Exam Checklist

Congratulations! You have mastered the physics of steady, moving charges.

Before advancing to Phase 5 (Time-Varying Fields and Maxwell's final unification), ensure you can confidently check off every item on this list:

- ☐ I can derive the Continuity Equation ($\nabla \cdot \mathbf{J} = -\partial \rho_v / \partial t$) by applying the Divergence Theorem to the macroscopic charge conservation integral.
- ☐ I can prove that the divergence of steady current is zero ($\nabla \cdot \mathbf{J} = 0$) because charge density cannot accumulate or deplete over time.
- ☐ I can establish Ampere's Law in point form ($\nabla \times \mathbf{H} = \mathbf{J}$) by applying Stokes' theorem to the integral form.
- ☐ I can define the Magnetic Vector Potential ($\mathbf{A}$) via the Coulomb Gauge, and I can follow the vector calculus derivation (using Poisson's equation) to deduce the Biot-Savart Law.
- ☐ I can confidently calculate the magnetic field of a square loop using Biot-Savart by systematically breaking it into 4 distinct line segments and calculating the cross products without sign errors.
- ☐ I can use a Gaussian pillbox and an Amperian loop to prove that normal $\mathbf{B}$ is continuous, while tangential $\mathbf{H}$ is discontinuous by the surface current $\mathbf{K}$.

Once this checklist is complete, you are fully prepared to tackle the final summit of the course: **Phase 5: Time-Varying Fields & EM Waves**.